COMPARISON WITH PUBLISHED MODELS

Advantages over Published Models

From lab accuracy to a field-ready, trustworthy diagnosis system

CropSense AI project | Technical report | 23 October 2026

1. Summary

Question. Does our model have any advantage over existing plant-disease models available online and in the literature?

Answer: yes, one major difference and several supporting ones. Published PlantVillage models are evaluated almost exclusively on **clean laboratory images** and report 99.35–99.97% accuracy there. Our model matches that level (99.56%) under a stricter, leakage-free protocol, and in addition is **measured and engineered for the field**: robust to blur, noise and poor light, with calibrated confidence and a filter that refuses to diagnose non-leaf images. The cited papers report none of these properties.

The major difference in one sentence: published models answer "which of 38 classes is this lab photo?"; CropSense answers "is this a single real leaf, what disease does it have, how sure are we, and where is the evidence?", and it stays accurate when the photo is blurred, noisy or dark.

Criterion	Published models (as reported)	CropSense: EfficientNetV2-MSAF + field aug.
Clean accuracy	99.35–99.97% (random file splits)	99.56% on a leaf-disjoint split: on par
Test-set leakage controlled	Not described in the papers	Yes: every copy of a leaf in one split (67.5% of a random split leaks)
Robustness to field corruptions	Not reported	94.8% mean over 6 corruptions; 81.8% under unseen motion blur
Calibrated confidence (ECE)	Not reported	5.5% (a stock EfficientNetV2-S trained identically: 10.6%)
Rejects non-leaf / multi-leaf images	No: always outputs a class	Yes: 5 of 6 non-leaf test images rejected
Real field photos	Mohanty et al.: 31.4% on other-source images	8/8 correct (small sample)
Explanation and action	Class label	Grad-CAM heatmap, severity estimate, treatment plan, PDF report

Published figures are quoted from the papers' abstracts and were not re-measured by us; "not reported" means the cited abstracts give no such result. Section 9 lists what this comparison cannot claim.

2. What published models report

The table below collects widely cited results for PlantVillage-style data, from the 2016 paper that introduced deep learning on PlantVillage to recent EfficientNet and DenseNet studies.

Study	Model	Data	Reported accuracy	Real-world test	Ref.
Mohanty et al., 2016	AlexNet / GoogLeNet	PlantVillage, 38 classes, 54,306 images	99.35%	31.4% on images from other online sources	[1]
Ferentinos, 2018	VGG and others	87,848 images, 58 classes	99.53%	Not reported	[2]
Too et al., 2019	DenseNet-121 (best of VGG16, Inception-V4, ResNet-50/101/152)	PlantVillage, 38 classes	99.75%	Not reported	[3]
Atila et al., 2021	EfficientNet-B4 / B5	PlantVillage (55,448 / 61,486 images)	99.91% / 99.97%	Not reported	[4]
Andrew et al., 2022	DenseNet-121	PlantVillage	99.81%	Not reported	[5]

Two observations from the literature matter for everything that follows:

- **Clean accuracy is saturated.** Every study is between 99.35% and 99.97%. A new model cannot show a meaningful gain there, so a real improvement has to be shown on other criteria.
- **Lab accuracy does not transfer to the field.** The only study in the table that tested on outside images, Mohanty et al. [1], reports a drop from 99.35% to **31.4%**. Noyan [6] later showed that a model using only **8 background pixels** of PlantVillage images reaches 49.0% accuracy (chance: 2.6%), so part of the high lab accuracy comes from background bias, not from the disease.

3. Advantage 1: a leakage-free evaluation

The Kaggle dataset used by many online implementations ships several rotated / flipped copies of each photo. If files are split at random, copies of the same physical leaf appear in both training and test sets, so the test partly measures memorisation.

- Our audit of a random file split: **67.5%** of test images had a rotated or flipped sibling in train/val.
- Our evaluation groups every copy of a leaf into one split (80/10/10 by leaf). The papers in Section 2 do not describe such grouping in their abstracts, so we cannot say whether their numbers are affected; ours are not.
- Under this stricter protocol our model still reaches **99.56%** on 8,889 test images, the same level as the published results.

4. Advantage 2: measured robustness to field conditions (the major difference)

Phone photos taken in a field are blurred by hand shake, noisy at dusk, dark in shade and compressed by messaging apps. We test six such corruptions, following the ImageNet-C methodology [8]. Four of them (motion blur, low light, JPEG, overexposure) were **never used in training**, so they measure genuine generalisation.

Condition	Stock EfficientNetV2-S*	CropSense (MSAF + field aug.)	Seen in training?
Gaussian blur	82.6%	**99.2%**	Similar
Motion blur	54.5%	**81.8%**	No
Sensor noise	80.7%	**99.4%**	Similar
Low light	90.0%	**96.8%**	No
JPEG (q=10)	86.7%	**93.6%**	No
Overexposure	98.8%	98.2%	No
Mean (mCA)	82.2%	**94.8%**	

**A standard EfficientNetV2-S, the same family as the EfficientNet models in [4], trained by us under the identical recipe and split. It stands in for a typical published model, because the cited papers give no corruption results.*

- Mean accuracy under corruption rises from **82.2% to 94.8%**; motion blur, never trained on, from 54.5% to 81.8%.
- This is the property the literature identifies as missing: lab accuracy that collapses outside the lab [1].

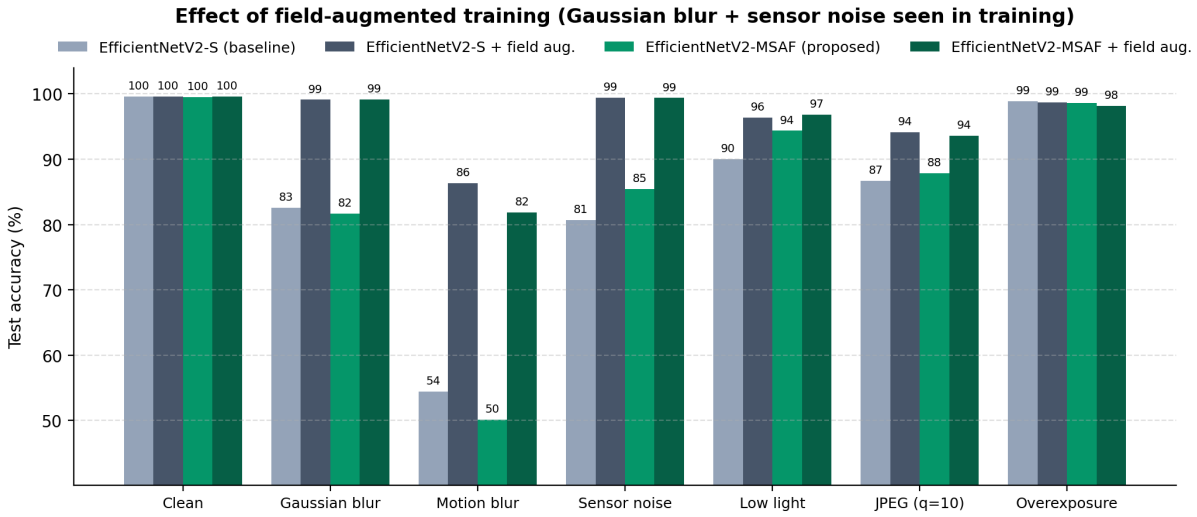


Figure 1. Accuracy per condition. The stock model collapses under motion blur and noise; the field-trained models do not.

5. Advantage 3: confidence you can trust

A diagnosis tool shows its confidence to farmers, and our app warns below 75%. That only works if a reported 90% is right about 90% of the time. This is measured by the Expected Calibration Error (ECE) [9]: lower is better.

Model (same split and recipe)	ECE	Field photos correct	Mean true-class confidence on field photos
Stock EfficientNetV2-S	10.57%	6/8	71%
EfficientNetV2-MSAF	5.14%	8/8	80%
EfficientNetV2-MSAF + field aug. (CropSense)	5.45%	8/8	76%

The MSAF architecture roughly **halves the calibration error** of a standard EfficientNetV2-S. None of the papers in Section 2 report calibration.

6. Advantage 4: it refuses to diagnose what is not a leaf

A standard classifier is "closed-world": whatever it receives, it outputs one of its 38 classes. CropSense first runs a leaf gatekeeper (plant-pigment colour analysis plus contour analysis) and rejects non-plant or multi-leaf photos.

We tested it on six non-leaf images:

Input image	Bare classifier (what a closed-world model says)	CropSense pipeline
Text document	Corn (maize): healthy (81%)	**Rejected: not a plant**
App screenshot	Apple: Black rot (37%)	**Rejected: not a plant**
Blank grey surface	Peach: healthy (15%)	**Rejected: not a plant**
Random noise	Tomato: Tomato mosaic virus (18%)	Passed, 18% confidence -> low-confidence warning
Blue sky	Soybean: healthy (70%)	**Rejected: not a plant**
Four-leaf collage	Pepper bell: healthy (49%)	**Rejected: multiple leaves**

Without the gatekeeper the classifier calls a **text document "Corn (maize): healthy" with 81% confidence**. CropSense rejects 5 of 6; the remaining one (random noise) passes the colour filter but gets a low confidence that triggers the app's warning. The test images are synthetic or composed by us, so this is a functional check rather than a benchmark (`scripts/gatekeeper_test.py`).

7. Advantage 5: explanation and action, not just a label

Output	Typical published model	CropSense
Disease class	Yes	Yes, top-3 with confidence
Visual evidence	No	Grad-CAM heatmap over the leaf
How badly is the leaf affected?	No	Severity estimate (share of leaf flagged, 4 stages)
What should the farmer do?	No	Pathogen, symptoms, organic and chemical control, prevention
Shareable record	No	One-click PDF diagnostic report
Field input	Offline images	Upload or live phone / webcam camera

Severity is a heuristic estimate from Grad-CAM, not a laboratory measurement; the app labels it as such.

8. Model size compared with published architectures

Architecture	Parameters (approx.)	Used in
VGG-16	138.4M	[2], [3]
AlexNet	61.1M	[1]
Inception-V4	42.7M	[3]
EfficientNet-B5	30.4M	[4]
ResNet-50	25.6M	[3]
EfficientNetV2-MSAF (ours)	20.4M	GPU latency 17 ms / image (RTX 3050 Laptop)
EfficientNet-B4	19.3M	[4]
DenseNet-121	8.0M	[3], [5]
GoogLeNet	6.8M	[1]

At 20.4M parameters our model is about 7× smaller than VGG-16 and smaller than ResNet-50 and EfficientNet-B5, but larger than DenseNet-121 and GoogLeNet. Size is not our main advantage; it is competitive, not leading.

9. What this comparison does not claim

Stating the limits keeps the advantages defensible in a viva.

- **No head-to-head re-run of published models.** Their numbers are quoted, not re-measured. The robustness and calibration comparison uses a stock EfficientNetV2-S trained by us as a representative standard model.
- **Clean accuracy is not an advantage.** Ours (99.56%) is on par with, and slightly below, the best reported results (99.75–99.97%), which were measured on random splits.
- **The field-photo test is small** (8 photos) and different from Mohanty et al.'s set, so 8/8 vs. 31.4% is an indication, not a controlled comparison. A larger field benchmark (e.g. PlantDoc [7]) is the natural next step.
- **Background bias** [6] is not specifically removed by our model; field augmentation and the gatekeeper reduce, but do not eliminate, reliance on lab-style backgrounds.
- **One training run per model**; differences under about 1 point are within noise.

10. How to present this to the faculty

"Published PlantVillage models reach 99.35–99.97% on clean lab images, but the original study shows accuracy falling to 31.4% on outside images, and none of these papers measure robustness, calibration or non-leaf inputs. Our system reaches the same clean accuracy (99.56%) on a stricter, leakage-free split, keeps 94.8% accuracy across six field corruptions (motion blur 81.8% without ever training on it), halves the calibration error of a standard EfficientNetV2-S, rejects non-leaf images that a closed-world classifier would confidently mislabel, and explains every diagnosis. Our major difference is field readiness and trustworthiness, not lab accuracy."

References

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